

Classification and/or Beyond

This is the final deliverable of your project. You will extend your work from P1 and P2 to tackle either a **classification task** or an **advanced regression / unsupervised task**, apply multiple models, perform systematic model selection, and produce a polished written report. The goal is to demonstrate that you can execute a complete, real-world ML pipeline from data to deployment-ready model.

Task Options

Choose **one** of the following tracks. State your choice at the top of your notebook.

Track A - Classification

Define a meaningful **binary or multi-class classification target** from your dataset. Examples:

- Binarize a continuous target (e.g., price above/below median)
- Use a naturally categorical column you set aside in P2
- Introduce a new column you engineer or label

Track B - Advanced Regression

If a classification framing does not make sense for your problem, stay with regression but add:

- At least one tree-based model (Decision Tree, Random Forest, or Gradient Boosting)
- Feature importance analysis
- Comparison against your P2 best model

Track C - Unsupervised Extension

Complement your supervised model with:

- K-Means or hierarchical clustering to discover natural groups
- PCA for dimensionality reduction and visualization
- Analysis of whether clusters align with the prediction target

Required Sections

1. Recap & Task Definition

- One-paragraph summary of your problem and dataset
- Define the classification target: what does each class mean?
- Report class distribution; if severely imbalanced, explain how you handle it

2. Data Preparation

- Reuse (and extend if needed) the feature engineering from P2
- Ensure train/validation/test split is consistent with P2 - do **not** re-split randomly

- Apply any additional preprocessing required by your chosen models (e.g., encoding, scaling)

3. Unsupervised Analysis

Apply unsupervised methods to understand the structure of your data **before** fitting supervised models. Fit all unsupervised models on the **training split only** to prevent data leakage.

3.1 Dimensionality Reduction with PCA

- Fit PCA on the training features (after scaling)
- Plot the **cumulative explained variance ratio** vs. number of components
- Report how many components are needed to explain at least 90 % of variance
- Visualize the data in **2D** using the first two principal components, coloured by the target label
- Discuss what the PCA projection reveals about the separability of classes

3.2 Clustering

Apply at least **two** of the following algorithms:

Method	Key Parameter(s) to Tune
K-Means	k - use the Elbow curve and Silhouette Score
Agglomerative (Hierarchical)	linkage criterion; cut dendrogram to choose k
DBSCAN	ϵ , <code>min_samples</code>
Gaussian Mixture Model (GMM)	<code>n_components</code> , covariance type

For each clustering method:

- Tune the key hyperparameter(s) and justify your final choice
- Visualize the resulting clusters in **2D PCA space**
- Compute the **Silhouette Score** and **Davies-Bouldin Index**
- Cross-tabulate or visualize cluster assignments against the target label - do the clusters align?

3.3 Cluster Labels as a Feature

Add the cluster assignment (from your best clustering run) as an additional engineered feature to your supervised pipeline. Evaluate whether this new feature improves classification performance and report the difference.

4. Model Training

Train **at least four** models. You must include **at least one ensemble method** from the table below:

Model	Key Hyperparameter(s) to Tune
Logistic Regression	C, penalty
K-Nearest Neighbours	k, distance metric
Decision Tree	max_depth, min_samples_leaf
Naive Bayes	-
Linear Discriminant Analysis	-
Support Vector Machine	C, kernel, γ
Random Forest	n_estimators, max_depth, max_features
Gradient Boosting	n_estimators, learning_rate, max_depth
Bagging Classifier	n_estimators, base estimator
Stacking Classifier	base estimators, meta-learner

For each model:

- Train on the training set
- Report validation-set performance: accuracy, precision, recall, F1 (macro)
- Produce a **confusion matrix**

5. Hyperparameter Tuning

- Use **Grid Search with cross-validation** for all models
- Report the best hyperparameters found and the cross-validation score
- Explain why you chose the hyperparameter grid you searched over

6. Model Comparison

Summarise all models in a single table:

Model	Val Accuracy	Val F1 (macro)	Test Accuracy	Test F1 (macro)	AUC-ROC
Logistic Regression					
Model 2					
Model 3					
Ensemble Model					
Best tuned model					

- Plot **performance curves** for all models on a single figure
- Select your **best model** and justify the choice

7. Error Analysis

- Examine misclassified examples: are there patterns?
- Which class is the model most confused about?
- Feature importance or model coefficients: which features drive predictions?

8. Final Written Report

Write a **structured report (600–1000 words)** inside the notebook using Markdown cells. It must include:

1. **Introduction** - problem, motivation, and dataset overview
2. **Methods** - what you did across P1, P2, and P3 (pipeline summary)
3. **Results** - key metrics and findings from all three deliverables
4. **Discussion** - what works, what doesn't, limitations, and potential improvements
5. **Conclusion** - summary of what you learned and the practical value of your model

Submission Requirements

Item	Details
File	Single Jupyter Notebook (.ipynb)
Filename	p3_classification_<student_id>.ipynb

GitHub Repository

- Push your notebook to the [p3/](#) folder of your existing repository
- Update [README.md](#) to reflect the completed project:
 - Final problem description
 - Dataset source
 - Short summary of findings across P1, P2, and P3
 - Instructions for running the notebooks

Submit the **link to your GitHub repository** on the course portal.